1.4.5 Training and simulation

Modern digital twins may provide immersive, virtual reality environments where teams can participate in active learning (particularly safe practice in hazardous environments), which can lead to better learning outcomes at a lower cost. The original purpose of the lunar and command module simulators setup by NASA during the Apollo space program was to provide a replica of the physical system which astronauts performed extensive training and simulation in.

REAL WORLD EXAMPLE: ANATOMICAL DIGITAL TWIN

Researchers at Curtin University's medical school in Western Australia used photogrammetry to produce 3D reconstructions of human anatomical specimens. These 3D models, as a digital representation of internal organs, decrease the need for anatomy students to have access to costly laboratory facilities with increasingly scarce physical resources, where they may also be at risk of exposure to chemicals as they study physical specimens.

1.4.6 Digital twins across industries

modeling the real world in software so that you can observe, understand, perform experiments, and predict future outcomes without changing the physical system is valuable and has advantages that apply to just about any industry that exists today, but we find digital twins mostly being applied in some key industries.

MINING, ENERGY AND INDUSTRIAL

Mining, energy production, and steel and metal processing are all examples of industries that are characterized by large-scale, capital-intensive operations that use complex processes to extract and transform raw materials into the essential components of our modern life. Within these industries, even small efficiency improvements to complex workflows and large operational costs can translate to significant financial, safety, and environmental outcomes. Digital twins contribute to process optimization in such industries by combining and contextualizing disparate process data and